



TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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FIRST REVISION EXAMINATION, JANUARY - 2025

Time Allowed : 3.00 Hours]

PHYSICS

[Max. Marks : 70

PART-I

I. Choose the correct answer.

15x1=15

- For light incident from air on a slab of Refractive index 2, the maximum possible angle of refraction is
a) 30° b) 45° c) 60° d) 90°
- If voltage applied on a capacitor is increased from 1V to 2V. Choose the Correct conclusion.
a) Q remains the same, C is doubled b) Q is doubled, C doubled
c) C remains same, Q doubled d) Both Q and C remains same
- When a positively charged conductor is earth connected
a) protons flow from the conductor to the earth
b) electrons flow from the earth to the conductor
c) electrons flow from the conductor to the earth d) no charge flow occurs
- The internal resistance of a 2.1 V cell which gives a current of 0.2 A through a resistance of $10\ \Omega$ is
a) $0.2\ \Omega$ b) $0.5\ \Omega$ c) $0.8\ \Omega$ d) $1.0\ \Omega$
- At Magnetic Poles the angle of Dip is
a) 45° b) 30° c) 0° d) 90°
- In a series RL circuit, the resistance and inductive reactance are the same. Then the phase difference between the voltage and current in the circuit is
a) $\pi/4$ b) $\pi/2$ c) $\pi/6$ d) Zero
- Which of the following has maximum Frequency?
a) X-rays b) IR Rays c) uv. Rays d) Radio Waves
- A carbon resistance has colour bands in order yellow, brown, red. Its resistance is
a) $41\ \Omega$ b) $4.1 \times 10^2\ \Omega$ c) $4 \times 10^3\ \Omega$ d) $4.2\ \Omega$
- A toaster operating at 240V has a resistance of $120\ \Omega$. The power is
a) 400 W b) 2 W c) 480 W d) 240 W
- A circular coil of radius 5 cm and has 50 turns carries a current of 3 ampere. The magnetic dipole moment of the coil is nearly
a) 1.0 Am^2 b) 1.2 Am^2 c) 0.5 Am^2 d) 0.8 Am^2
- Eddy currents are produced in a material when it is
a) Heated b) Placed in a time varying magnetic field
c) Placed in an electric field d) Placed in a uniform magnetic field
- Fraunhofer lines are an example of ----- spectrum
a) Line Emission b) line Absorption c) Band Emission d) band absorption
- Dispersion of light is caused due to
a) Wavelength b) intensity of light c) density of medium d) none of these
- One henry (H) is equal to
a) $\text{V s}^{-1} \text{A}^{-1}$ b) VsA^{-1} c) $\text{V}^{-1}\text{s}^{-1}\text{A}^{-1}$ d) $\text{V}^{-1} \text{sA}^{-1}$
- The Speed of light in an isotropic medium depends on,
a) Its Intensity b) Its Wavelength
c) The nature of Propagation d) The motion of the Source w.r.t medium



PART - II

II. Answer Any SIX of The Following. (Answer Question No.24 Compulsory)

6x2=12

16. Define Electrostatic Potential.
17. Two cells each of 5V are connected in series across a 8Ω resistor and three parallel resistors of 4Ω , 6Ω and 12Ω . Draw a circuit diagram for the above arrangement.
18. State Lenz's Law?
19. What is Snell's window.
20. What are the application of seebeck effect?
21. Write down the equation for a sinusoidal voltage of 50 Hz and its peak value is 20 V.
22. What is meant by Hysteresis?
23. What are uses of UV Radiation?
24. Calculate the distance upto which ray optics is a good approximation for light of wavelength 500nm falls on an aperture of width 0.5 mm?

PART- III

III. Answer Any SIX of The Following. (Answer Question No.33 Compulsory)

6x3=18

25. Obtain an expression for electric fields due to two parallel charged infinite sheets using Gauss's law
26. Discuss the Hertz Experiment?
27. A coil of a tangent galvanometer of diameter 0.24 m has 100 turns. If the horizontal component of Earth's magnetic field is $25 \times 10^{-6} \text{ T}$ then, calculate the current which gives a deflection of 60° .
28. State Kirchhoff's Rules.
29. What are advantages and Disadvantages of AC over DC?
30. Consider a point charge $+q$ placed at the origin and another point charge $-2q$ placed at a distance of 9 m from the charge $+q$. Determine the point between the two charges at which electric potential is zero.
31. Write note on Nicol Prism.
32. Give the properties of ferromagnetic materials.
33. Two air core solenoids have the same length of 80 cm and same cross-sectional area 5 cm^2 . Find the mutual inductance between them if the number of turns in the first coil is 1200 turns and that in the second coil is 400 turns.

PART - IV

IV. Answer ALL Questions.

5x5=25

34. a) Show mathematically that the rotation of a coil in a magnetic field over one rotation induces an alternating emf of one cycle. (OR)
b) Explain the construction and working of Transformer.
35. a) Describe the microscopic model of current and obtain Microscopic form of Ohm's law. (OR)
b) i) Write down any Six Properties of Electromagnetic Waves.
ii) If the relative permeability and relative permittivity of the medium are 1.0 and 2.25, respectively. Find the speed of the electromagnetic wave in this medium.
36. a) Obtain Lens Maker's formula. (OR)
b) What is Dispersion? obtain the Equation for Dispersive power of a medium
37. a) Discuss the working of Cyclotron in Detail. (OR)
b) i) State Ampere's Circuital law.
ii) Find the Magnetic Field due to a Long straight conductor using Ampere's circuital law
38. a) Explain in detail the construction and Working of a Van de Graaff generator. (OR)
b) Explain in detail the effect of a Dielectric placed in a Parallel plate Capacitor, when the Capacitor is disconnected from the battery.